

CS422: Computer Architecture Benchmark Analysis

Team Members:

Mrityunjay Kumar (230668)

Krishna Agrawal (230574)

Sarthak Verma (230931)

Krish Jain (230572)

Contents

1	Implementation Details	2
2	400.perlbench	3
3	401.bzip2	5
4	403.gcc	7
5	429.mcf	9
6	450.soplex	11
7	456.hmmcr	13
8	471.omnetpp	15
9	483.xalancbmk	17
10	Observations	19

1 Implementation Details

Overview

This PIN tool instruments 32-bit ia32 SPEC 2006 benchmark binaries to collect dynamic instruction statistics (Part A), estimate CPI (Part B), measure memory footprint (Part C), and analyze ISA properties (Part D). The tool is built using Pin 4.1 and compiled as a 32-bit shared library (`obj-ia32/HW1.so`) targeting the ia32 ISA.

Fast-Forward and Termination

Each benchmark is fast-forwarded over a SimPoint-determined number of instructions and then analyzed for exactly **1 billion instructions**. A global counter `icount` is incremented for every instruction, including those with false predicates.

Instrumentation Structure

Instrumentation is performed at the **TRACE level**, iterating over BBLs and then individual instructions.

To avoid a slow switch-case at analysis time, **type-A category counters are resolved at instrumentation time**. A pointer to the appropriate global counter is passed as an `IARG_PTR` argument to the single analysis routine `mem_count`, which increments that counter and adds loads/stores in one call.

Part B — CPI Estimation

CPI is calculated directly from Part A counters using:

$$\text{CPI} = \frac{(\text{numLoads} + \text{numStores}) \times 70 + (\text{total} - \text{numLoads} - \text{numStores}) \times 1}{\text{total}}$$

Build and Usage

Build:

```
make TARGET=ia32 obj-ia32/HW1.so
```

On a 32-bit host, omit `TARGET=ia32`. Debug symbols can be enabled with `DEBUG=1`.

Run:

```
cd /path/to/spec_2006/<benchmark>/
/path/to/pin -t /path/to/obj-ia32/HW1.so \
  -f <ff_billions> -o <output_file> -- ./<binary> <args>
```

The `-f` knob specifies the fast-forward amount in billions of instructions and `-o` specifies the output file. All results are written to that file before `exit(0)` is called.

2 400.perlbench

Number of instructions: 208000000006
Fast forward at: 207000000000
Number of instructions after fast forward: 1000000006

=====PARTA=====

Number of loads:	356094197 (22.79%)
Number of stores:	206367704 (13.21%)
Number of nops:	979403 (0.06%)
Number of direct calls:	12875494 (0.82%)
Number of indirect calls:	2805529 (0.18%)
Number of returns:	15681028 (1.00%)
Number of unconditional branches:	30489651 (1.95%)
Number of conditional branches:	130079969 (8.33%)
Number of logical operations:	100394909 (6.43%)
Number of rotate/shift operations:	4272179 (0.27%)
Number of flag operations:	870570 (0.06%)
Number of vector operations:	0 (0.00%)
Number of conditional moves:	0 (0.00%)
Number of MMX/SSE operations:	0 (0.00%)
Number of system calls:	0 (0.00%)
Number of FP operations:	960632 (0.06%)
Number of other instructions:	700426099 (44.83%)

=====PARTB=====

CPI: 25.84

Part C Analysis Results:

Instruction Footprint Blocks	= 2808
Instruction Footprint (in bytes)	= 89856
Data Footprint Blocks	= 31250
Data Footprint (in bytes)	= 1000000
Single Memory Chunk Instruction accesses	= 921647980
Multiple Memory Chunk Instruction accesses	= 78352024
Single Memory Chunk Data accesses	= 552312173
Multiple Memory Chunk Data accesses	= 25099

=====

Part D Analysis Results:

Distribution of instruction length Results:

1 : 117701414
2 : 256866328
3 : 274902520

4 : 53001487
5 : 78606088
6 : 184969038
7 : 33953089
8 : 28
10 : 12

Distribution of number of operands Results:

0 : 978824
1 : 1072984
2 : 520375553
3 : 354956126
4 : 103684606
5 : 15688073
6 : 3243838

Instructions with 2 register read operands : 537970090
Instructions with 1 register write operand : 684850154
Instructions with 3 memory operands : 0
Instructions with 1 memory read operand : 354358936
Instructions with 2 memory write operands : 0
Max bytes touched by a memory instruction : 8
Avg bytes touched by a memory instruction : 3.739345
Max immediate value : 2147483647
Min immediate value : -2147483648
Max displacement value : 135918104
Min displacement value : -1408

3 401.bzip2

Number of instructions: 302000000010
Fast forward at: 301000000000
Number of instructions after fast forward: 1000000010

=====PARTA=====

Number of loads:	452705283 (26.88%)
Number of stores:	231174532 (13.73%)
Number of nops:	36514 (0.00%)
Number of direct calls:	791544 (0.05%)
Number of indirect calls:	18 (0.00%)
Number of returns:	791559 (0.05%)
Number of unconditional branches:	21299366 (1.26%)
Number of conditional branches:	129922994 (7.72%)
Number of logical operations:	71001190 (4.22%)
Number of rotate/shift operations:	61832454 (3.67%)
Number of flag operations:	6130 (0.00%)
Number of vector operations:	0 (0.00%)
Number of conditional moves:	0 (0.00%)
Number of MMX/SSE operations:	0 (0.00%)
Number of system calls:	0 (0.00%)
Number of FP operations:	0 (0.00%)
Number of other instructions:	714314162 (42.42%)

=====PARTB=====

CPI: 29.02

=====
Part C Analysis Results:

Instruction Footprint Blocks = 753
Instruction Footprint (in bytes) = 24096
Data Footprint Blocks = 2441019
Data Footprint (in bytes) = 78112608
Single Memory Chunk Instruction accesses = 902089980
Multiple Memory Chunk Instruction accesses= 97910027
Single Memory Chunk Data accesses = 602693513
Multiple Memory Chunk Data accesses = 2
=====

Part D Analysis Results:

Distribution of instruction length Results:

1 : 38611273
2 : 219201470
3 : 436965694

4 : 75326977
5 : 22047451
6 : 141357282
7 : 51341618
8 : 15085632
10 : 62610

Distribution of number of operands Results:

0 : 36514
1 : 6147
2 : 597647499
3 : 382870391
4 : 2693851
5 : 14191397
6 : 2554208

Instructions with 2 register read operands : 533608181
Instructions with 1 register write operand : 712782411
Instructions with 3 memory operands : 0
Instructions with 1 memory read operand : 452706092
Instructions with 2 memory write operands : 0
Max bytes touched by a memory instruction : 8
Avg bytes touched by a memory instruction : 3.485407
Max immediate value : 1431655766
Min immediate value : -858993459
Max displacement value : 135000192
Min displacement value : -4848

4 403.gcc

Number of instructions: 1080000000000
Fast forward at: 1070000000000
Number of instructions after fast forward: 1000000000

=====PARTA=====

Number of loads:	338875033 (22.45%)
Number of stores:	170335288 (11.29%)
Number of nops:	3906167 (0.26%)
Number of direct calls:	30224718 (2.00%)
Number of indirect calls:	396033 (0.03%)
Number of returns:	30620751 (2.03%)
Number of unconditional branches:	31771889 (2.11%)
Number of conditional branches:	145812275 (9.66%)
Number of logical operations:	54593537 (3.62%)
Number of rotate/shift operations:	634876 (0.04%)
Number of flag operations:	0 (0.00%)
Number of vector operations:	0 (0.00%)
Number of conditional moves:	0 (0.00%)
Number of MMX/SSE operations:	0 (0.00%)
Number of system calls:	0 (0.00%)
Number of FP operations:	0 (0.00%)
Number of other instructions:	702039758 (46.52%)

=====PARTB=====

CPI: 24.28

=====
Part C Analysis Results:

Instruction Footprint Blocks = 2953
Instruction Footprint (in bytes) = 94496
Data Footprint Blocks = 1132231
Data Footprint (in bytes) = 36231392
Single Memory Chunk Instruction accesses = 977152232
Multiple Memory Chunk Instruction accesses= 22847769
Single Memory Chunk Data accesses = 485892653
Multiple Memory Chunk Data accesses = 0
=====

Part D Analysis Results:

Distribution of instruction length Results:

1 : 129712163
2 : 591194824
3 : 125305257

4 : 115815964
5 : 11303792
6 : 15794353
7 : 10815362
8 : 58286

Distribution of number of operands Results:

0 : 188515
1 : 4602210
2 : 349737513
3 : 403980536
4 : 33854080
5 : 203381864
6 : 4255283

Instructions with 2 register read operands : 472674528
Instructions with 1 register write operand : 419395219
Instructions with 3 memory operands : 0
Instructions with 1 memory read operand : 138733447
Instructions with 2 memory write operands : 0
Max bytes touched by a memory instruction : 8
Avg bytes touched by a memory instruction : 3.958460
Max immediate value : 1073741823
Min immediate value : -2147483587
Max displacement value : 138634432
Min displacement value : -1744

5 429.mcf

Number of instructions: 378000000002
Fast forward at: 377000000000
Number of instructions after fast forward: 1000000002

=====PARTA=====

Number of loads:	415211113 (27.22%)
Number of stores:	110041734 (7.21%)
Number of nops:	1477682 (0.10%)
Number of direct calls:	12555802 (0.82%)
Number of indirect calls:	0 (0.00%)
Number of returns:	12555801 (0.82%)
Number of unconditional branches:	8314673 (0.55%)
Number of conditional branches:	178242826 (11.69%)
Number of logical operations:	75119699 (4.93%)
Number of rotate/shift operations:	3516304 (0.23%)
Number of flag operations:	0 (0.00%)
Number of vector operations:	0 (0.00%)
Number of conditional moves:	0 (0.00%)
Number of MMX/SSE operations:	0 (0.00%)
Number of system calls:	0 (0.00%)
Number of FP operations:	0 (0.00%)
Number of other instructions:	708217216 (46.43%)

=====PARTB=====

CPI: 24.76

=====

Part C Analysis Results:

Instruction Footprint Blocks	= 65
Instruction Footprint (in bytes)	= 2080
Data Footprint Blocks	= 11672851
Data Footprint (in bytes)	= 373531232
Single Memory Chunk Instruction accesses	= 946574123
Multiple Memory Chunk Instruction accesses	= 53425879
Single Memory Chunk Data accesses	= 512817670
Multiple Memory Chunk Data accesses	= 0

=====

Part D Analysis Results:

Distribution of instruction length Results:

1 : 80626424
2 : 485393431

3 : 315526365
4 : 50531222
5 : 22076562
6 : 5249816
7 : 40596182

Distribution of number of operands Results:

0 : 1477639
2 : 484784790
3 : 457347075
4 : 43834189
5 : 12556309

Instructions with 2 register read operands : 677530613
Instructions with 1 register write operand : 770283684
Instructions with 3 memory operands : 0
Instructions with 1 memory read operand : 415212722
Instructions with 2 memory write operands : 0
Max bytes touched by a memory instruction : 4
Avg bytes touched by a memory instruction : 4.000000
Max immediate value : 1374389535
Min immediate value : -100000000
Max displacement value : 134957120
Min displacement value : -76

6 450.soplex

Number of instructions: 365000000004
Fast forward at: 364000000000
Number of instructions after fast forward: 1000000004

=====PARTA=====

Number of loads:	546776100 (33.20%)
Number of stores:	100104807 (6.08%)
Number of nops:	9406 (0.00%)
Number of direct calls:	3194808 (0.19%)
Number of indirect calls:	126 (0.00%)
Number of returns:	3194928 (0.19%)
Number of unconditional branches:	12975966 (0.79%)
Number of conditional branches:	103136818 (6.26%)
Number of logical operations:	13878363 (0.84%)
Number of rotate/shift operations:	10418428 (0.63%)
Number of flag operations:	23123509 (1.40%)
Number of vector operations:	0 (0.00%)
Number of conditional moves:	0 (0.00%)
Number of MMX/SSE operations:	0 (0.00%)
Number of system calls:	0 (0.00%)
Number of FP operations:	307966306 (18.70%)
Number of other instructions:	522101355 (31.70%)

=====PARTB=====

CPI: 28.10

=====
Part C Analysis Results:

Instruction Footprint Blocks = 645
Instruction Footprint (in bytes) = 20640
Data Footprint Blocks = 5658598
Data Footprint (in bytes) = 181075136
Single Memory Chunk Instruction accesses = 936203952
Multiple Memory Chunk Instruction accesses= 63796054
Single Memory Chunk Data accesses = 463714579
Multiple Memory Chunk Data accesses = 4806452
=====

Part D Analysis Results:

Distribution of instruction length Results:

1 : 77591064
2 : 441944549
3 : 399698035

4 : 16087417
5 : 3219650
6 : 40002784
7 : 17519595
8 : 3936912

Distribution of number of operands Results:

0 : 3550
1 : 13
2 : 412391769
3 : 396045279
4 : 188351008
5 : 3208387

Instructions with 2 register read operands : 578652926
Instructions with 1 register write operand : 765092931
Instructions with 3 memory operands : 0
Instructions with 1 memory read operand : 416818852
Instructions with 2 memory write operands : 0
Max bytes touched by a memory instruction : 8
Avg bytes touched by a memory instruction : 5.270663
Max immediate value : 2147483647
Min immediate value : -640172613
Max displacement value : 135855532
Min displacement value : -344

7 456.hmmmer

Number of instructions: 2650000000006
Fast forward at: 2640000000000
Number of instructions after fast forward: 10000000006

=====PARTA=====

Number of loads:	547670100 (33.74%)
Number of stores:	75698505 (4.66%)
Number of nops:	34317 (0.00%)
Number of direct calls:	144622 (0.01%)
Number of indirect calls:	959 (0.00%)
Number of returns:	145581 (0.01%)
Number of unconditional branches:	205862 (0.01%)
Number of conditional branches:	144361650 (8.89%)
Number of logical operations:	1158706 (0.07%)
Number of rotate/shift operations:	294108 (0.02%)
Number of flag operations:	5669 (0.00%)
Number of vector operations:	0 (0.00%)
Number of conditional moves:	0 (0.00%)
Number of MMX/SSE operations:	0 (0.00%)
Number of system calls:	0 (0.00%)
Number of FP operations:	40212 (0.00%)
Number of other instructions:	853607360 (52.58%)

=====PARTB=====

CPI: 27.50

Part C Analysis Results:

Instruction Footprint Blocks	= 485
Instruction Footprint (in bytes)	= 15520
Data Footprint Blocks	= 84052
Data Footprint (in bytes)	= 2689664
Single Memory Chunk Instruction accesses	= 927512713
Multiple Memory Chunk Instruction accesses	= 72487288
Single Memory Chunk Data accesses	= 623213681
Multiple Memory Chunk Data accesses	= 1000

=====

Part D Analysis Results:

Distribution of instruction length Results:

1 : 25078459
2 : 302860553

3 : 296150913
4 : 270388798
5 : 24861234
6 : 68360208
7 : 416480
8 : 11883356

Distribution of number of operands Results:

0 : 34887
1 : 3512
2 : 566126761
3 : 432938367
4 : 713115
5 : 159551
6 : 23808

Instructions with 2 register read operands : 562398802
Instructions with 1 register write operand : 755280001
Instructions with 3 memory operands : 0
Instructions with 1 memory read operand : 547662878
Instructions with 2 memory write operands : 0
Max bytes touched by a memory instruction : 8
Avg bytes touched by a memory instruction : 3.997511
Max immediate value : 2147483647
Min immediate value : -987654321
Max displacement value : 135294312
Min displacement value : -580

8 471.omnetpp

Number of instructions: 44000000014
Fast forward at: 43000000000
Number of instructions after fast forward: 1000000014

=====PARTA=====

Number of loads:	371203570 (23.19%)
Number of stores:	229800698 (14.35%)
Number of nops:	802287 (0.05%)
Number of direct calls:	21333563 (1.33%)
Number of indirect calls:	3688430 (0.23%)
Number of returns:	25021990 (1.56%)
Number of unconditional branches:	22188346 (1.39%)
Number of conditional branches:	117336461 (7.33%)
Number of logical operations:	60021492 (3.75%)
Number of rotate/shift operations:	7138166 (0.45%)
Number of flag operations:	20157260 (1.26%)
Number of vector operations:	0 (0.00%)
Number of conditional moves:	0 (0.00%)
Number of MMX/SSE operations:	0 (0.00%)
Number of system calls:	0 (0.00%)
Number of FP operations:	96954812 (6.06%)
Number of other instructions:	625357211 (39.06%)

=====PARTB=====

CPI: 26.90

=====

Part C Analysis Results:

Instruction Footprint Blocks	= 890
Instruction Footprint (in bytes)	= 28480
Data Footprint Blocks	= 528736
Data Footprint (in bytes)	= 16919552
Single Memory Chunk Instruction accesses	= 958067988
Multiple Memory Chunk Instruction accesses	= 41932013
Single Memory Chunk Data accesses	= 551105073
Multiple Memory Chunk Data accesses	= 4470346

=====

Part D Analysis Results:

Distribution of instruction length Results:

1 : 154578809
2 : 308551256

3 : 382074562
4 : 34347877
5 : 45117271
6 : 48822846
7 : 26506959
10 : 421

Distribution of number of operands Results:

0 : 802403
1 : 225879
2 : 518294355
3 : 281743917
4 : 172763957
5 : 25030575
6 : 1138915

Instructions with 2 register read operands : 540431957
Instructions with 1 register write operand : 665542419
Instructions with 3 memory operands : 0
Instructions with 1 memory read operand : 337160174
Instructions with 2 memory write operands : 0
Max bytes touched by a memory instruction : 8
Avg bytes touched by a memory instruction : 4.221750
Max immediate value : 2147483647
Min immediate value : -2092037281
Max displacement value : 136090116
Min displacement value : -104

9 483.xalancbmk

Number of instructions: 1332000000002
Fast forward at: 1331000000000
Number of instructions after fast forward: 1000000002

=====PARTA=====

Number of loads:	365059037 (23.85%)
Number of stores:	166074868 (10.85%)
Number of nops:	23142145 (1.51%)
Number of direct calls:	12784974 (0.84%)
Number of indirect calls:	8929411 (0.58%)
Number of returns:	21715081 (1.42%)
Number of unconditional branches:	8365380 (0.55%)
Number of conditional branches:	181363010 (11.85%)
Number of logical operations:	36943566 (2.41%)
Number of rotate/shift operations:	5448615 (0.36%)
Number of flag operations:	1610174 (0.11%)
Number of vector operations:	0 (0.00%)
Number of conditional moves:	0 (0.00%)
Number of MMX/SSE operations:	0 (0.00%)
Number of system calls:	0 (0.00%)
Number of FP operations:	7029442 (0.46%)
Number of other instructions:	692210439 (45.22%)

=====PARTB=====

CPI: 24.94

=====
Part C Analysis Results:

Instruction Footprint Blocks = 2293
Instruction Footprint (in bytes) = 73376
Data Footprint Blocks = 835433
Data Footprint (in bytes) = 26733856
Single Memory Chunk Instruction accesses = 968811611
Multiple Memory Chunk Instruction accesses= 31188391
Single Memory Chunk Data accesses = 529617896
Multiple Memory Chunk Data accesses = 1929077
=====

Part D Analysis Results:

Distribution of instruction length Results:

1 : 145803853
2 : 320583305
3 : 445456155

4 : 32054791
5 : 24198387
6 : 23775448
7 : 7735564
8 : 262712
9 : 61599
10 : 68188

Distribution of number of operands Results:

0 : 22013641
1 : 751907
2 : 422984354
3 : 411595170
4 : 102360929
5 : 24276033
6 : 16017968

Instructions with 2 register read operands : 456830833
Instructions with 1 register write operand : 683913578
Instructions with 3 memory operands : 0
Instructions with 1 memory read operand : 365574934
Instructions with 2 memory write operands : 0
Max bytes touched by a memory instruction : 8
Avg bytes touched by a memory instruction : 4.161971
Max immediate value : 2147483647
Min immediate value : -1431655765
Max displacement value : 139657912
Min displacement value : -1392

10 Observations

1. The high CPI values (24–29) across all benchmarks reflect the dominance of memory operations and their 70-cycle penalty. `401.bzip2` has the highest CPI (29.02), consistent with its heavy load (26.88%) and store (13.73%) fractions, while `403.gcc` has the lowest (24.28).
2. `429.mcf` has the smallest instruction footprint (65 blocks, 2 KB) reflecting a tight inner loop, but the largest data footprint (~373 MB) owing to its large graph traversal working set. `401.bzip2` shows a large data footprint (78 MB) relative to its small instruction footprint (24 KB), consistent with block-compression data access patterns.
3. The majority of instructions are 2–3 bytes across all benchmarks, reflecting common short-form ia32 encodings. A small number of 8–10 byte instructions appear in floating-point-heavy benchmarks such as `450.soplex`.
4. Most instructions have 2 or 3 operands, consistent with standard x86 encoding forms. Instructions with 0 operands are primarily NOPs; those with 5–6 operands are typically complex string or SIMD operations.
5. Instructions with 2 register read operands represent the most common binary arithmetic pattern, while 1 register write operand is the typical result-update model.
6. No benchmark exhibited instructions with 3 memory operands or 2 memory write operands, consistent with the ia32 constraint of at most one explicit memory destination in standard ALU instructions.
7. The maximum bytes touched by a single memory instruction is 8 across all benchmarks (64-bit double-precision loads/stores), except `429.mcf` where the maximum is 4 bytes. The average bytes touched ranges from 3.49 (`401.bzip2`) to 5.27 (`450.soplex`), the latter reflecting its heavy use of 8-byte floating-point operands.
8. Maximum displacements are consistently in the 134–139 MB range across benchmarks, corresponding to addresses in high regions of the 32-bit address space. Minimum displacements are small negative values (e.g., `-4848` in `401.bzip2`), representing stack-relative accesses.